



Chenhuan(Choy) Cai

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Summary

Research Interests: AI Algorithm Engineer, Computer Vision Engineer | Solid theoretical foundation and project experience in Machine Learning, Computer Vision, 3D Reconstruction, and Multimodal Modeling.

Skills: Python, C++, Linux | Git, Bash, Conda, CUDA, HPC, Slurm | PyTorch, Hugging Face, Transformers, Diffusers, etc.

Education

University of Zurich

M.Sc. in Artificial Intelligence GPA: 5.25 / 6.00

2023.09 – 2027.01

Zurich, Switzerland

Core Courses: Foundations of Data Science(6.00), Deep Learning, Computer Vision, Digital Humans, 3D Vision(5.75), etc.

Harbin Institute of Technology, Shenzhen

B.Sc. in Data Science GPA: 3.7 / 4.0

2019.09 – 2023.06

Shenzhen, China

Core Courses: Data Structures, Advanced Algebra, Probability Theory, Differential Equations, Statistical Machine Learning, etc.

Research & Internship

ETH Zurich

2025.05 – 2025.12

3D Vision Research, Computer Vision and Geometry Group (CVG)

Zurich, Switzerland

- Worked on novel view synthesis for 3D reconstruction, where existing methods such as NoPoSplat suffer from limited robustness under low-overlap views and strong dependence on reference frames.
- Proposed *GenSplat*, combining diffusion-driven generative data augmentation with a permutation-equivariant Gaussian fusion framework. The method improves robustness in low-overlap scenarios and achieves better LPIPS, PSNR, SSIM, and qualitative results than NoPoSplat on RE10K and DL3DV.

NetEase Youdao Information Technology (Beijing) Co., Ltd.

2022.04 – 2022.06

Computer Vision Algorithm Engineer, AI Product Department

Beijing, China

- Participated in the R&D of visual detection technology for the NetEase Youdao Aladdin smart desk lamp, responsible for developing and evaluating the finger-paper contact detection module.
- Leveraged pseudo-labeling, image processing, data augmentation, enhancement, and tracking techniques. Based on YOLOX, the project achieved stable detection in lights-on scenarios and improved lights-off accuracy to over 80%, validating the feasibility of the contact detection solution and supporting subsequent product deployment.

McGill University

2022.07 – 2022.09

Machine Learning Research Assistant, Division of Experimental Medicine

Remote

- Participated in a bioinformatics project, building a web-based pipeline for data preprocessing, dimensionality reduction, clustering, and visualization, and performed single-cell data analysis using Scanpy and Velocity.

Projects & Publications

ShapeNav: 3D Geometric Perception and Reasoning for Multi-Floor VLN.

2024.09 – 2025.05

Wu, Y.*, Cai, C.*, Yan, L., Li, M., Xu, C., Yang, C., Guan, F. & Li, P.

Under review.

- Addressed the lack of explicit 3D geometric perception in robotic navigation by proposing ShapeNav, a framework that integrates point clouds with vision-language models.
- Designed Geometric Patch Extraction (GPE) and Masked Optimal Transport (MOT) modules to align 3D shape features with 2D visual features.

STaR: Multi-Granular Spatio-Temporal Reasoning for Long-Form Dense Video Caption.

2024.04 – 2024.08

Wu, Y.*, Cai, C.*, Li, H., Bai, Z. & Yan, L.

ECAI 2025.

- Addressed insufficient long-context understanding and spatial information degradation in dense video captioning by proposing STaR, a multi-granular spatio-temporal reasoning framework.
- Designed a Context Capture (CC) module to fuse global and short-window local features, and a Spatial Semantic Synthesis (SSS) module that enhances spatial awareness through hash encoding.

Awards & Honors

Kaggle: Stable Diffusion - Image to Prompts | Silver Medal

2023.02 – 2023.05

- Built training data from images generated by Stable Diffusion and aligned image-text embeddings through contrastive learning to predict prompts from images.

National Undergraduate Mathematical Contest in Modeling | National Second Prize

2021.09

Teddy Cup Data Mining Challenge | National Third Prize

2021.05

First-Class Academic Scholarship, Outstanding Student Award, First Prize in Mathematics Competition

2020